The Silent Divides in Education's Promise: Uneven Wealth Gains from College

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Abstract

College is widely seen as a path to wealth, yet large racial wealth gaps persist among graduates. Using PSID data from 1999 to 2023, this paper examines why. Random-effects and within-sibling estimates show large wealth gains from college for White graduates but small or negative gains for Non-White graduates, even though earnings returns are similar. Student debt is the main channel: Non-White graduates are nearly twice as likely to hold it, and excluding it from net wealth narrows the racial gap by about two-thirds and eliminates it within sibling pairs. Debt financing thus limits college's wealth returns for Non-White households.

Keywords: Racial wealth gap · Returns to education · Student debt · Household wealth · Intergenerational transfers

JEL Codes: I24 · I26 · J15

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1 Introduction

Education is associated with higher earnings (Card, 1999) and with substantial financial returns to college degrees (Cappelli, 2020), and recent evidence suggests that educational policies can raise wealth accumulation (Loaiza, 2024). Yet large racial wealth gaps remain among college graduates. If college equalized opportunities, graduates of all racial backgrounds would convert their degrees into similar wealth gains. The persistence of gaps among equally educated households suggests that the wealth returns to education depend on household financial conditions. This study asks whether a college degree is associated with similar wealth gains across racial groups, and which channels account for the differences.

The distinction between earnings and wealth is central to this question. Education may raise labor income, but wealth also depends on debts, transfers, and the assets a household is able to acquire. Equal earnings gains need not imply equal wealth gains if households differ in inherited resources, in how much they borrow, or in their access to appreciating assets. For this reason, the analysis looks beyond earnings to parental wealth, inheritances, and student debt, and compares siblings raised in the same family.

Figure 1 shows the puzzle. College is associated with similar earnings gains for White and Non-White individuals: median labor income is about $33,000 higher for White college graduates than for White non-graduates, and about $36,000 higher among Non-White individuals. The corresponding wealth gains are very different. Median net wealth is about $92,000 higher for White college graduates than for White non-graduates, but only about $15,500 higher among Non-White individuals. These differences appear before adjusting for family background, life-cycle differences, or other covariates.

A large literature documents that racial wealth gaps are rooted in unequal access to housing and credit markets, institutional barriers to asset accumulation, and persistent differences in intergenerational transfers (Derenoncourt et al., 2024; Chetty et al., 2020). Recent Survey of Consumer Finances evidence shows that wealth remains sharply stratified by race and education, even as median net worth rose between 2019 and 2022

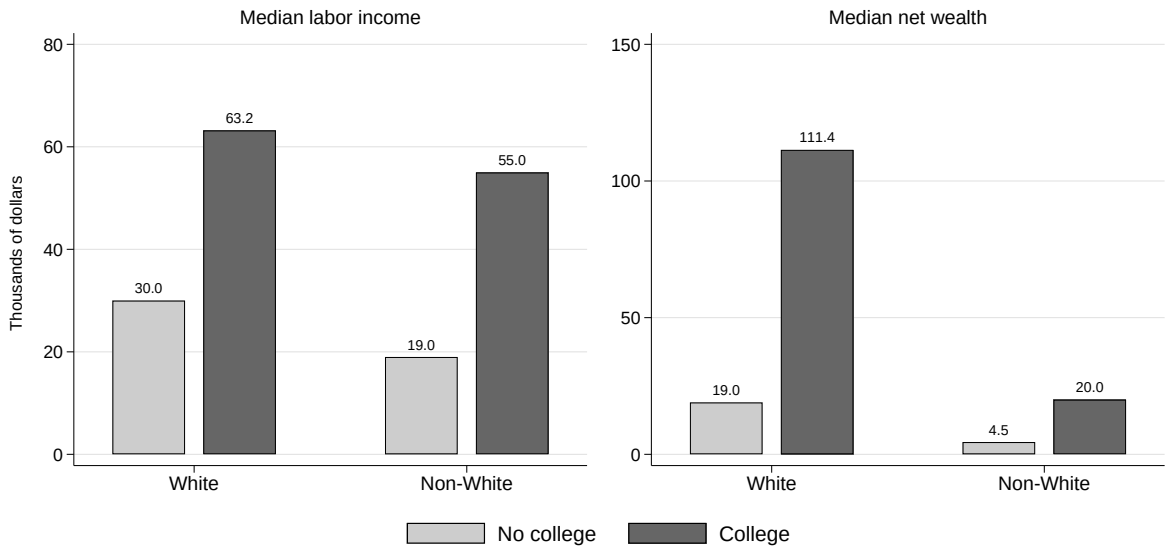


Figure 1: Income and Wealth by Race and College

Notes: Source: PSID, 1999–2023, baseline estimation sample (Table 1). Weighted medians in thousands of nominal dollars. Net wealth excludes home equity.

(Aladangady et al., 2023). Parental wealth, inheritances, and family resources are especially important because they shape both educational opportunities and later asset accumulation (Pfeffer & Killewald, 2019). White families are more likely to receive financial support for college, homeownership, and business formation, while non-White families have historically faced greater constraints in transferring wealth across generations (Charles & Hurst, 2003; Fagereng et al., 2021; Taylor & Meschede, 2018).

Education influences wealth directly through labor income and indirectly through financial literacy, investment behavior, and portfolio composition. College graduates generally hold more wealth, yet racial gaps persist even among the college educated (Bartscher, Kuhn, & Schularick, 2020; Meschede, Taylor, Mann, & Shapiro, 2017; Emmons & Ricketts, 2017). More educated individuals tend to exhibit greater financial sophistication and hold higher-yield financial assets (Lusardi & Mitchell, 2023; Loaiza, 2024, 2025). At the same time, student debt and unequal access to high-return assets may weaken the link between education and wealth for non-White households (Houle & Addo, 2019; Addo et al., 2016).

This study uses data from the Panel Study of Income Dynamics (PSID) from 1999 to 2023 and centers the empirical analysis on a within-sibling design. Comparing biolog-

ical siblings who grow up in the same household but attain different levels of education isolates variation in educational attainment net of shared family background. Baseline random-effects regressions provide descriptive benchmarks, while an instrumental variables approach based on parental job loss is reported only as supplementary evidence.

The results show large racial differences. White college graduates hold substantially more wealth than White non-graduates, whereas the gain for Non-White graduates is small and sometimes negative. The difference persists when siblings raised in the same family are compared, and it does not come from the labor market, since college raises labor income by similar amounts for both groups. The evidence instead points to how college is financed. Non-White graduates borrow more often, and once student debt is set aside, their wealth gains from college are positive and much closer to those of White graduates. Parental wealth remains a strong predictor of adult wealth, and the gap that remains suggests that intergenerational transfers also matter.

This paper makes three contributions. First, it provides new evidence on racial heterogeneity in the wealth returns associated with tertiary education, shifting attention from average education premia to how college attainment translates into household balance sheets. Second, it accounts for family background using parental wealth itself rather than a proxy. Studies of the returns to education typically capture family resources through parental income or education, which miss much of the wealth that parents can transfer to their children. The intergenerational structure of the PSID makes it possible to measure parental wealth directly, before children reach adulthood, and the sibling design further absorbs all background shared within the family. Third, by comparing net wealth with and without student debt, it identifies student debt as the main channel through which the wealth return to college differs by race, and it shows that the earnings return does not differ. The central finding is that college raises earnings and gross assets for both groups, but Non-White graduates finance it with more debt, which offsets much of its wealth payoff.

The remainder of the paper proceeds as follows. Section 2 describes the data and empirical strategy. Section 3 reports the main findings. Section 4 examines descriptive

mechanisms, and Section 5 discusses implications and concludes.

2 Data and Empirical Strategy

2.1 Data and Measures

This study uses the biennial waves of the Panel Study of Income Dynamics (PSID) from 1999 to 2023. The PSID follows families over time and allows intergenerational links between parents and adult children. The analysis uses two samples. The first is a baseline adult panel with complete information on wealth, education, and covariates. The second is a restricted sample of biological siblings, used for within-family comparisons.

The unit of observation in the baseline analysis is an adult individual observed in a PSID survey wave. Individuals enter the sample once they are older than age 30 and are coded as the household head or reference person. This restriction focuses the analysis on adult household wealth after most formal schooling is complete and avoids mixing adult children's own household wealth with wealth reported for the parental household. All specifications include age, year, and birth-cohort controls.

The outcome is household net wealth in the adult respondent's current household, not wealth in the family of origin. Wealth includes financial assets, retirement accounts, business and farm equity, real estate holdings, and mortgage and non-mortgage debt. Social Security wealth and defined-benefit pensions are excluded. Because wealth can be zero or negative, the analysis uses a generalized inverse hyperbolic sine transformation with scaling parameter $\lambda = 0.0001$. Appendix Section A1 provides the formula and interpretation. This transformation is defined at zero and for negative wealth while behaving similarly to a logarithm for large positive values. Coefficients are changes in transformed wealth rather than in dollars; Appendix Table A1 converts the main estimates into dollars.

The key explanatory variable is an indicator for tertiary education, equal to one for individuals who completed a four-year college degree or postgraduate education. Baseline models include parental wealth, parental education, parental presence at age 16, inheritances, age, race, sex, and proxies for individual ability and school performance.

In the sibling specifications, shared family characteristics are absorbed by the family comparison.

Parent-child links are constructed using PSID family identifiers. Parental characteristics are measured from the respondent's family of origin and merged onto the adult child panel. Parental wealth is taken from available pre-adult household wealth measures and assigned by birth cohort, so that family resources are measured before or near the transition into adulthood. Parental education is measured separately for the household head and spouse, and parental presence indicates whether the respondent lived with both parents at age 16. Race is measured using the PSID race variable and coded as White versus Non-White. This aggregation is used because the sibling design requires sufficient sample size within family-linked observations. The category combines heterogeneous groups and should not be interpreted as a single social or economic experience. Mechanisms may differ across Black, Hispanic, Asian, Indigenous, and other racial or ethnic groups.

Table 1 describes the baseline estimation sample by race and college attainment. The wealth differences are large at every level of education: the median net wealth of White college graduates is more than five times that of Non-White college graduates. The table also anticipates the main results. Non-White college graduates are more likely to have negative net wealth than Non-White individuals without a degree (27 compared with 19 percent), whereas among White individuals the share is similar with and without a degree. Non-White graduates are also nearly twice as likely as White graduates to hold student debt (44 compared with 24 percent in 2011–2023), even though their median labor income is not far below that of White graduates ($55,000 compared with $63,200). Family resources differ sharply as well. The median parental wealth of White college graduates is about seven times that of Non-White college graduates, and White graduates are twice as likely to have received an inheritance.

2.2 Empirical Strategy

The analysis begins with baseline random-effects regressions:

Table 1: Descriptive Statistics

	White		Non-White	
	No college	College	No college	College
Wealth				
Median net wealth ($)	19,000	111,400	4,500	20,000
Net wealth ≤ 0 (%)	20.2	14.7	33.1	29.0
Net wealth < 0 (%)	16.1	14.2	18.8	27.2
Labor market				
Median labor income ($)	30,000	63,200	19,000	55,000
Any labor income (%)	76.0	86.5	69.9	89.0
Student debt (2011–2023)				
Holds student debt (%)	13.5	23.7	16.6	44.5
Median balance if any ($)	16,000	36,000	15,000	30,000
Family background				
Median parental wealth ($)	25,000	59,000	4,000	8,350
Received an inheritance (%)	4.6	8.1	1.4	3.9
Both parents at age 16 (%)	89.3	92.4	81.8	84.7
Demographics				
Mean age	49.1	49.0	48.9	46.1
Female (%)	31.9	24.5	46.5	36.5
Person-years	9,739	6,520	7,362	1,424
Individuals	1,492	1,007	1,121	286

Note: Source: PSID, 1999–2023. Baseline estimation sample: household heads or reference persons older than 30 with complete information on wealth, education, and covariates. Statistics are weighted; counts are unweighted. Dollar values are nominal; net wealth excludes home equity. Student-debt rows use 2011–2023.

$$Y_{it} = \beta_0 + \beta_1 \text{Education}_i + \beta_2 X_i + \beta_3 D_{it} + \gamma_t + c_i + \upsilon_{it}, \tag{1}$$

where Y_{it} denotes household wealth, Education_i indicates tertiary education, X_i includes time-invariant parental and individual characteristics, D_{it} includes time-varying adult covariates, and γ_t captures year fixed effects. These models provide descriptive evidence on the education–wealth association and on the importance of family resources.

The main research design uses within-sibling variation in educational attainment. The sibling file is constructed from PSID family links by matching biological siblings into pairwise sibling records. Each observation in the sibling specifications is a sibling-pair-by-year comparison in which both siblings are observed in the same survey wave, are older than age 30, and are coded as household heads or reference persons. Families with more than two observed siblings can contribute more than one pairwise comparison. In the main pooled sibling interaction sample, the estimating sample contains 7,870 pair-year

observations, corresponding to 1,273 unique sibling pairs. Identification comes from pairs whose college attainment differs: about 20 percent of pair-years are discordant in college attainment, and 321 pairs (25 percent) are discordant in at least one wave, 195 White and 126 Non-White pairs. Appendix Table A2 reports this variation by race, and Appendix Table A3 shows that siblings differ in broad educational attainment in about half of all pairs.

$$\Delta Y_{pt} = \alpha_0 + \alpha_1 \Delta \text{Education}_{pt} + \alpha_2 \Delta X_{pt} + \gamma_t + \upsilon_{pt}, \tag{2}$$

where p indexes sibling pairs, t indexes survey years, and Δ denotes the difference between siblings observed in the same wave. This specification removes characteristics shared by the pair and includes within-pair differences in age, childhood socioeconomic conditions, school performance, and behavioral indicators. Because siblings can appear in repeated waves and larger families can contribute multiple pairs, the main sibling tables cluster standard errors at the sibling-pair level. The design reduces confounding from shared family background, but it does not rule out all within-family differences in parental investments, peer networks, aspirations, or individual-specific endowments.

As supplementary evidence, Appendix Table A9 reports instrumental-variable estimates that use parental job loss when the child is between ages 15 and 18 as an instrument for college attainment. Because parental job loss may affect later wealth directly through household resources and transfers, and because the instrument is weak for Non-White individuals, these estimates are not used for inference.

3 Results

Table 2 reports the main estimates. Columns (1) and (2) report baseline random-effects estimates separately for White and Non-White individuals, columns (3) and (4) report within-sibling estimates by race, and column (5) pools White and Non-White sibling pairs with an interaction between tertiary education and Non-White status. Column (5) provides the most direct test of the central hypothesis, because it tests whether the within-

family education–wealth association differs by race rather than relying on comparisons across separately estimated subgroup regressions.

Table 2: Wealth Returns to College by Race

Dependent Variable: Wealth					
	Baseline RE		**Within-Sibling**		
	White (1)	Non-White (2)	White (3)	Non-White (4)	Pooled (5)
Tertiary	3428.81***	−3623.54***	5529.08*	−661.14	5445.82*
	(796.46)	(1064.55)	(2249.76)	(1837.00)	(2237.89)
Tertiary × Non-White					−5700.88*
					(2846.78)
Inheritance	0.24***	0.36***			
	(0.02)	(0.06)			
Parental presence	1303.12	466.59			
	(1040.20)	(716.93)			
Parental wealth	0.29***	0.13***			
	(0.03)	(0.04)			
Mother's education	462.70	192.04			
	(320.65)	(354.22)			
Father's education	400.92	853.25+			
	(333.05)	(435.49)			
Observations	16259	8888	4177	3693	7870
R^2	0.24	0.11	0.03	0.02	0.02

Note: Source: PSID, 1999–2023. Robust standard errors in columns (1)–(2); standard errors clustered by sibling pair in columns (3)–(5). $^{+}$ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Columns (1)–(2) are random-effects models that also control for age, sex, ability, year, and birth cohort. Columns (3)–(5) use sibling differences and control for differences in age, childhood conditions, school performance, and behavior; family background shared by siblings is absorbed. Column (5) pools White and Non-White pairs; Tertiary is the White return and Tertiary × Non-White the difference. R^2: overall in (1)–(2), adjusted in (3)–(5).

In the baseline models, tertiary education is associated with large and statistically significant wealth gains for White individuals, whereas the estimate for Non-White individuals is negative and statistically significant. The within-sibling estimates show the same pattern: a significant gain for White siblings and a small, statistically insignificant estimate for Non-White siblings. In the pooled sibling model, the Tertiary × Non-White interaction is negative, similar in magnitude to the White coefficient, and statistically significant when standard errors are clustered at the sibling-pair level. The implied within-family wealth return for Non-White sibling pairs is therefore close to zero. Because the within-sibling specifications remove characteristics shared by siblings, these results indicate that unequal wealth returns to college are not solely attributable to differences

in shared family background, although the precision of the interaction is reduced once repeated pair observations are accounted for.

In dollar terms, these differences are large relative to the wealth of non-graduates (Appendix Table A1). Evaluated at the median net wealth of White non-graduates ($19,000), the baseline estimate implies that a college degree is associated with about $8,600 more net wealth for White individuals, and the within-sibling estimate implies about $15,000. Evaluated at the median net wealth of Non-White non-graduates ($4,500), the corresponding estimates for Non-White individuals are about −$3,800 in the baseline model and close to zero (−$300) within sibling pairs.

These patterns are robust to alternative treatments of the wealth distribution (Appendix Table A8). When wealth is winsorized or trimmed at the 1st and 99th percentiles, the baseline estimates remain positive and significant for White individuals and negative for Non-White individuals, while the within-sibling estimates remain significant for White sibling pairs and close to zero for Non-White pairs. The log specification, which drops zero and negative wealth, yields positive coefficients for both groups. This suggests that the weaker returns for Non-White graduates are driven largely by the lower part of the wealth distribution, where net wealth is low, zero, or negative.

3.1 Family Resources and Intergenerational Wealth

Family resources shape wealth accumulation across generations, so the education estimates are reported alongside them. A large literature emphasizes that parental wealth, inheritances, inter vivos transfers, and family background affect adult wealth through channels that extend beyond educational attainment (De Nardi, 2004; De Nardi & Yang, 2016; Karagiannaki, 2017; Black et al., 2020; Fagereng et al., 2020). Recent life-cycle evidence from Norwegian tax records further shows that parental wealth is central to persistent wealth positions, while human capital is more closely linked to mobility through the middle of the distribution (Audoly et al., 2024). Columns (1) and (2) of Table 2 report these variables directly in the baseline random-effects specification.

Family resources are strongly associated with adult wealth. Inheritance is positive

and statistically significant for both groups, and parental wealth strongly predicts adult wealth in both. These resources may also shape whether a degree turns into lasting wealth.

Panel (a) of Figure 2 shows how parental wealth and college attainment combine. Adult wealth rises with parental wealth for every group, and the college premium is largest among individuals from wealthier families. This pattern could suggest that Non-White graduates would obtain similar wealth returns if their parents were similarly wealthy. The figure does not support that reading. In the bottom half of the parental wealth distribution, Non-White college graduates hold only modestly more wealth than Non-White individuals without a degree (median wealth of about \$9,000–\$12,500 compared with about \$3,000–\$4,500), while White college graduates in the second quartile already hold about \$60,000. In the top quartile, Non-White college graduates have a median net wealth of about \$145,000, compared with about \$220,000 for White college graduates with parents in the same quartile. Parental wealth raises the wealth of Non-White graduates, but it does not close the gap with White graduates from similarly wealthy families. This comparison should be read with care. Few Non-White individuals have parents in the top quartile (122 person-years among Non-White college graduates), and because the quartiles are broad, Non-White parents within a given quartile may still hold less wealth than White parents in the same quartile.

Panel (b) of Figure 2 plots the college wealth premium directly, defined as the difference in median net wealth between college graduates and non-graduates within each group and parental wealth quartile. The premium is smaller for Non-White individuals in every quartile: about \$4,000 compared with \$9,000 in the bottom quartile, \$9,000 compared with \$40,000 in the second, \$47,000 compared with \$87,000 in the third, and \$134,000 compared with \$178,000 in the top quartile. The gap is widest in the middle of the parental wealth distribution, where the Non-White premium is less than a quarter of the White premium in the second quartile and about half in the third. Even in the top quartile, the college wealth premium for Non-White individuals is only about three-quarters of the premium for White individuals with parents in the same quartile.

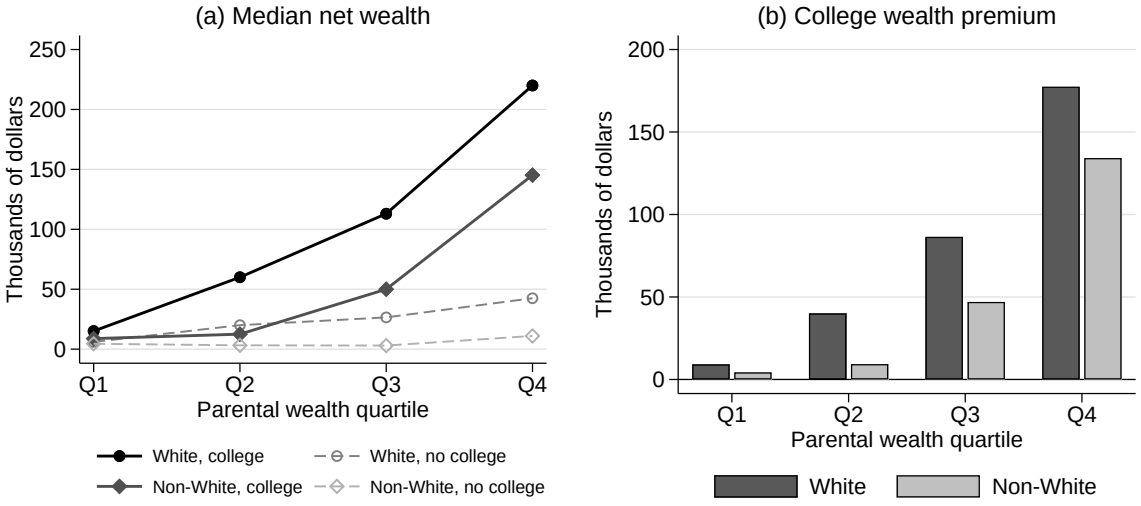


Figure 2: Wealth and the College Premium by Parental Wealth

Notes: Source: PSID, 1999–2023, baseline estimation sample. Panel (a): weighted median net wealth by quartile of parental wealth. Panel (b): difference in median net wealth between college graduates and non-graduates. Thousands of dollars.

In the within-sibling specifications, these family-background variables are absorbed by the family comparison, because siblings share the same parental wealth, parental education, and early-life environment. This is why they do not appear in the sibling columns of Table 2. Their coefficients in the baseline models are associations and should not be read causally.

3.2 Additional Heterogeneity

Appendix Section A4 examines whether the college wealth premium changed across broad birth cohorts, comparing individuals born in 1939–58 and 1959–88 at ages 40–50. Appendix Table A7 tests cohort differences directly in pooled models with a Tertiary × Non-White × cohort interaction. In the random-effects models, the college wealth premium for White individuals is similar across cohorts, and the difference is not statistically significant. For Non-White individuals, the premium declines sharply, from about 7,800 for the earlier cohort to about −4,500 for the later cohort, and the triple interaction shows that this decline is significantly larger than for White individuals. When the comparison is restricted to 2001–2009, the years in which both cohorts are observed at ages 40–50, the

decline for Non-White individuals remains statistically significant but is smaller, and the racial difference in the cohort change is no longer significant. Part of the pattern therefore appears to reflect the later calendar period rather than birth cohort alone. That period includes the Great Recession and the housing crisis, which reduced Non-White wealth disproportionately. The within-sibling estimates are too imprecise to test cohort differences, since restricting siblings to ages 40–50 leaves few pairs with different educational attainment in each cohort. These cohort patterns are descriptive and consistent with evidence that the financial returns to college have fallen for more recent cohorts (Emmons, Kent, & Ricketts, 2019). They may also reflect changes in who attends college, in college quality and fields of study, and in student debt, rather than a decline in the value of a given degree.

4 Mechanisms

This section examines two descriptive channels, labor income and student loan debt, and then compares net wealth with and without student debt. The analysis estimates the total education–wealth association, the association between education and each mediator, and the wealth regression with the mediator included. It then reports pooled race-interaction tests that directly assess whether these gradients differ by race. Both the race-specific models in the appendix and the pooled interaction models in Table 3 are random-effects specifications with standard errors clustered at the individual level. Neither uses the within-sibling design, because differencing removes much of the variation in mediators. Student loans are observed from 2011 onward, so all student-loan models use the 2011–2023 waves. The results describe correlated channels and should not be read as causal mediation.

4.1 Labor Income as a Channel

Appendix Table A4 presents the race-specific labor-income estimates. Tertiary education is associated with higher labor income for both White and Non-White individuals. In

Table 3: Income and Student Debt Channels

	Wealth	Mediator	Wealth + Mediator
	(1)	(2)	(3)
Panel A: Labor income, full sample, 1999–2023			
Tertiary	4160.80***	4963.70***	3521.90***
	(754.00)	(413.20)	(747.50)
Non-White	−2887.10***	−2313.70***	−2565.30***
	(651.60)	(353.70)	(633.20)
Tertiary × Non-White	−8883.40***	560.80	−8948.80***
	(1259.70)	(666.40)	(1252.10)
Labor income			0.15***
			(0.02)
Observations	25147	25147	25147
Individuals	3660	3660	3660
Overall R^2	0.27	0.24	0.29
Panel B: Student loans, student-loan sample, 2011–2023			
Tertiary	5458.90***	3322.40***	9092.60***
	(901.70)	(388.70)	(788.40)
Non-White	−3369.30***	−320.80	−3826.80***
	(712.70)	(322.80)	(633.50)
Tertiary × Non-White	−10763.70***	5462.20***	−4757.30***
	(1431.30)	(821.90)	(1199.90)
Student loans			−1.01***
			(0.03)
Observations	15808	15808	15808
Individuals	3277	3277	3277
Overall R^2	0.29	0.17	0.44

Note: Source: PSID. Random-effects estimates pooling White and Non-White individuals; standard errors clustered by individual. $^{+}$ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Column (1): wealth; column (2): the mediator (labor income in Panel A, student loans in Panel B); column (3): wealth with the mediator added. Panel B uses the waves with observed student loans. Controls as in Table 2, columns (1)–(2).

Table 3, the Tertiary × Non-White coefficient in the labor-income equation is small and statistically insignificant, so the education–earnings gradient does not differ detectably by race. Including labor income in the wealth equation also does not attenuate the negative Tertiary × Non-White interaction. Education raises earnings for both groups, but the higher earnings do not translate into wealth to the same extent. The remaining gap points to constraints outside the labor market, such as family transfers and debt.

Appendix Table A6 shows that this conclusion is robust to alternative income measures. The Tertiary × Non-White interaction is small and statistically insignificant for log labor income among workers, and separately for men and women. In weighted descriptive statistics, median labor income rises from $30,000 to $63,000 with a college degree among

White individuals and from \$19,000 to \$55,000 among Non-White individuals. The only significant racial difference in the income equations runs in the opposite direction from the wealth gap: tertiary education raises the probability of having any labor income by about 4 percentage points more for Non-White individuals. At the very top of the distribution, however, mean income gains are larger for White graduates, which may contribute to differences in wealth accumulation among the highest earners. Overall, there is no evidence that the earnings return to college is smaller for Non-White individuals.

4.2 Student Loan Burden and Wealth Accumulation

Appendix Table A5 examines student loan debt in race-specific models. Student debt is negatively related to wealth, and the pooled interaction test shows that the education-debt gradient is substantially steeper for Non-White individuals. Within the 2011–2023 student-loan sample, the Tertiary $\times$ Non-White wealth interaction falls from $-10{,}764$ to $-4{,}757$ once student loans are included, a reduction of roughly 56 percent, but it remains statistically significant. In dollar terms, the implied college wealth effect for Non-White individuals in this sample moves from about $-\$5{,}400$ to about $+\$5{,}300$ once student loans are included (Appendix Table A1). The race-specific models show the same pattern on a common sample: the Non-White tertiary coefficient moves from $-4{,}684$ to a positive and statistically significant 5,005 once student loans are added. Student debt thus accounts for a substantial part of the racial difference, though not all of it. The student-loan coefficient in the wealth equation is close to -1, which is what one would expect if each dollar of student debt reduces net wealth by about one dollar. Because student loans enter net wealth directly and are observed for a smaller sample, this is an accounting decomposition rather than evidence of a behavioral mechanism.

Figure 3 illustrates both parts of this result. Among individuals without student debt, wealth rises steeply with education for both groups, although far more for White individuals. Among individuals with student debt, the education gradient largely disappears: median net wealth is close to zero or negative at most education levels. White college graduates with student debt are the exception, with a median net wealth of about

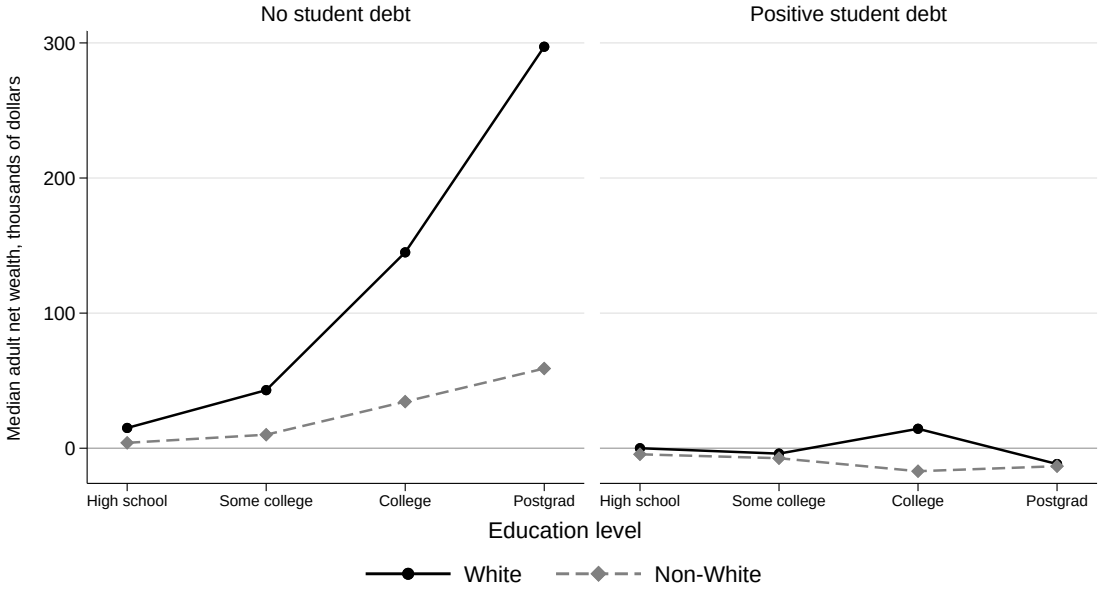


Figure 3: Wealth by Education and Student Debt

Notes: Source: PSID, 2011–2023, baseline estimation sample. Weighted median net wealth by education and by whether the household holds student debt. Thousands of dollars; not adjusted for age or other covariates.

$14,000, whereas Non-White college graduates with student debt have a median of about −$17,000. Non-White graduates are also much more likely to be in this second group: in the unweighted 2011–2023 estimation sample, 45 percent of Non-White college graduates and 76 percent of Non-White postgraduates report student debt, compared with 26 and 31 percent of White college graduates and postgraduates. Debt does not account for the whole gap, however. Even among debt-free individuals, the median net wealth of Non-White college graduates is about $35,000, compared with $145,000 for White college graduates; among debt-free postgraduates it is $59,000 compared with $297,000, although the Non-White estimate rests on only 116 person-years. Individuals with student debt are also younger on average, which the figure does not adjust for.

4.3 Wealth Excluding Student Debt

Because student debt is subtracted from net wealth, the student-loan results above partly reflect accounting rather than behavior. Table 4 therefore asks a more direct question: does college raise the wealth of Non-White graduates once student debt is excluded from

the balance sheet? The outcome adds the student-loan balance back to net wealth, and each comparison uses the same observations for both outcomes. The analysis is restricted to the waves with observed student-loan balances: 2011–2023 for the random-effects models and 2011–2019 for the sibling models, since student loans are not observed for sibling pairs in 2021 and 2023.

Table 4: Wealth Returns Excluding Student Debt

	Random Effects, 2011–2023		**Within-Sibling, 2011–2019**	
	Net Wealth	Excl. Student Debt	Net Wealth	Excl. Student Debt
Tertiary (White)	5458.86***	8547.36***	3516.36	8034.71***
	(901.73)	(735.68)	(2493.62)	(2375.87)
Tertiary × Non-White	−10763.73***	−3625.37***	−3919.32	−241.76
	(1431.34)	(1092.51)	(3303.07)	(2822.75)
Tertiary effect, Non-White	−5304.87***	4921.99***	−402.96	7792.95***
	(1160.98)	(861.66)	(2245.85)	(1616.50)
Observations	15808	15808	3641	3641

Note: Source: PSID, waves with observed student loans. Standard errors clustered by individual (random effects) or sibling pair (within-sibling; 1,012 pairs). $^{+}$ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Excluding student debt adds the student-loan balance back to net wealth. Both outcomes use the same observations within each design. Tertiary effect, Non-White is the sum of Tertiary and Tertiary × Non-White. Controls as in Tables 2 and 3.

The results point to student debt as the main reason Non-White graduates do not gain wealth from college in this period. With net wealth as the outcome, the college effect for Non-White individuals is negative in the random-effects model and zero within sibling pairs. Excluding student debt, it becomes positive and statistically significant in both designs: about \$6,200 at the median wealth of Non-White non-graduates in the random-effects model, and somewhat larger within sibling pairs. College raises the gross assets of Non-White graduates, but the debt incurred to finance it offsets those gains. In the random-effects model, the Tertiary × Non-White interaction shrinks by about two-thirds, from −10,764 to −3,625, and remains statistically significant. Within sibling pairs, it falls to −242 and is far from statistical significance: comparing siblings from the same family, the remaining racial difference in the wealth return to college is accounted for by student debt.

Student debt is a real liability, so Non-White graduates are no better off than their net wealth suggests. What the results show is that the racial gap in wealth returns operates

largely through how college is financed. In 2011–2023, 44 percent of Non-White college graduates hold student debt, compared with 24 percent of White graduates (Table 1). Student debt is not observed before 2011, so these results cannot speak to the earlier years of the sample. The gap that remains after excluding debt, together with the debt-free medians in Figure 3, indicates that differences in family wealth also contribute.

5 Discussion and Conclusion

Using PSID data from 1999 to 2023, this study shows that the wealth premium associated with tertiary education differs sharply by race. In the pooled sibling specification, the Tertiary × Non-White interaction is negative, similar in magnitude to the White education premium, and statistically significant. Race-specific estimates show the same pattern: White individuals exhibit substantial wealth gains associated with tertiary education, whereas Non-White individuals do not show comparable gains.

Parental wealth and inheritances remain strong predictors of adult wealth. The sibling design absorbs shared family background, so the sibling estimates answer a narrower question: whether education-related differences remain once that background is held fixed. They do not imply that family resources are unimportant.

College raises labor income by similar amounts for both groups, but only White graduates turn those gains into comparable wealth. In the years for which student debt is observed, debt is the main reason. Non-White graduates borrow more often, and setting their student debt aside removes most of the racial gap in the full sample and all of it within families. Some gap remains in the full sample, and student debt is not observed before 2011, so differences in family wealth also appear to matter.

Several limitations remain. Survey-reported wealth, inheritances, and transfers may contain measurement error, and the PSID lacks detailed information on portfolio allocation, risk preferences, and informal transfers. Although the sibling design reduces confounding from shared family background, unobserved within-family differences, such as parental investments or individual endowments, may still influence both education and

wealth accumulation (Barth, Papageorge, Thom, & Velásquez-Giraldo, 2022). Recent PSID-based evidence also emphasizes persistent heterogeneity in wealth returns and labor earnings over the life cycle, which this paper cannot fully separate from racial differences in the education–wealth gradient (Daminato & Pistaferri, 2024). The parental-job-loss IV is reported only as supplementary evidence.

Higher education supports economic advancement, but its wealth payoff differs across racial groups. College remains a good investment and a central part of mobility policy, and the results suggest that its wealth payoff depends on how it is financed and on family resources. Because Non-White students rely more on borrowing, policies that reduce the debt needed to complete college, such as need-based grants or income-driven repayment, may do more to narrow racial wealth gaps among graduates than policies that raise completion alone.

Data Availability Statement

The data used in this study are from the Panel Study of Income Dynamics (PSID), which is publicly available from the Institute for Social Research, University of Michigan (`https://psidonline.isr.umich.edu`). Code to construct the analysis sample and reproduce all results will be made available upon publication.

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Appendix

A1 IHS Transformation and Dollar Magnitudes

For a raw wealth value W, the transformed outcome is

$$\widetilde{W} = \frac{\sinh^{-1}(\lambda W)}{\lambda} = \frac{\log\left(\lambda W + \sqrt{1+\lambda^2 W^2}\right)}{\lambda}, \qquad \lambda = 0.0001. \tag{3}$$

This transformation is defined at zero and for negative wealth while behaving similarly to a logarithm for large positive values. Coefficients are changes in transformed wealth, not literal dollar changes. A coefficient β can be translated back into dollars at a chosen baseline value $\widetilde{W}_0$ using

$$\Delta W(\widetilde{W}_0, \beta) = \frac{\sinh\{\lambda(\widetilde{W}_0 + \beta)\} - \sinh(\lambda \widetilde{W}_0)}{\lambda}.$$

Table A1 applies this formula to the main estimates. The baseline is the weighted median net wealth of non-graduates of the same race in the baseline estimation sample: \$19,000 for White and \$4,500 for Non-White individuals. Because the mapping is monotonic, 95 percent confidence intervals are obtained by translating the bounds of the coefficient's confidence interval. The dollar values depend on the chosen baseline: the same coefficient implies a larger dollar change at higher baseline wealth.

Table A1: Dollar Magnitudes

	Coefficient	Dollar change	95% CI (dollars)
Baseline random effects (Table 2)			
White	3,429	8,635	[4,366; 13,579]
Non-White	−3,624	−3,762	[−5,854; −1,639]
Pooled within-sibling (Table 2, column 5)			
White	5,446	15,167	[2,386; 34,628]
Non-White (sum of coefficients)	−255	−278	[−3,983; 3,976]
Student-loan sample, pooled random effects (Table 3)			
White, without student loans	5,459	15,213	[9,416; 22,082]
Non-White, without student loans	−5,305	−5,446	[−7,776; −3,165]
White, with student loans	9,093	30,737	[23,461; 39,202]
Non-White, with student loans	4,335	5,334	[2,820; 8,208]

Note: Coefficients on the transformed wealth scale, converted to dollars at the median net wealth of non-graduates of the same race (\$19,000 White; \$4,500 Non-White). Confidence intervals convert the bounds of each coefficient's 95 percent interval.

A2 Within-Sibling Variation in Education

Table A2: College Variation Within Sibling Pairs

	All pairs	White pairs	Non-White pairs
Sibling pairs	1,273	686	587
Pair-years	7,870	4,177	3,693
Pair-years discordant in college (%)	19.7	23.1	15.8
Both siblings college graduates (%)	14.4	23.9	3.6
Neither sibling a college graduate (%)	65.9	53.0	80.6
Pairs ever discordant	321	195	126
Pairs ever discordant (%)	25.2	28.4	21.5

Note: Estimation sample of Table 2, column (5). A pair-year is discordant if exactly one sibling has a four-year college degree. Unweighted.

Table A3: Sibling Education Transitions

	Sibling 2				
Sibling 1	Less than HS	High school	Some college	College+	Total
Less than HS	87	55	33	11	186
High school	57	184	125	60	426
Some college	27	93	142	83	345
College+	12	37	71	189	309
Total	183	369	371	343	1,266

Note: One observation per sibling pair (latest wave with schooling observed for both; 1,266 of 1,273 pairs). Less than HS: under 12 years of schooling; high school: 12; some college: 13–15; College+: 16 or more.

A3 Mechanism Estimates by Race

Table A4: Labor Income Channel by Race

	White			Non-White		
	Wealth (1)	Labor Income (2)	Wealth (3)	Wealth (4)	Labor Income (5)	Wealth (6)
Tertiary	3428.81***	5115.20***	2824.40***	−3623.54***	4997.50***	−4482.50***
	(796.46)	(421.30)	(789.90)	(1064.55)	(550.80)	(1060.00)
Labor income			0.14***			0.17***
			(0.02)			(0.03)
Observations	16259	16259	16259	8888	8888	8888
R^2	0.24	0.20	0.26	0.11	0.20	0.14

Note: Source: PSID, 1999–2023. Random-effects estimates by race; robust standard errors. $^{+}$ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Columns (3) and (6) add labor income to the wealth equation. Controls as in Table 2, columns (1)–(2). Overall R^2.

Table A5: Student Loan Channel by Race

	White			Non-White		
	Wealth (1)	Student Loans (2)	Wealth (3)	Wealth (4)	Student Loans (5)	Wealth (6)
Tertiary	4814.33***	3513.60***	8605.00***	−4683.55***	8743.80***	5005.00***
	(950.83)	(408.90)	(840.80)	(1206.92)	(727.60)	(991.20)
Student loans			−0.97***			−1.04***
			(0.05)			(0.04)
Observations	10009	10009	10009	5799	5799	5799
R^2	0.26	0.11	0.41	0.11	0.27	0.34

Note: Source: PSID, 2011–2023, waves with observed student loans. Random-effects estimates by race; robust standard errors. $^{+}$ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Columns (3) and (6) add student loans to the wealth equation. Controls as in Table 2, columns (1)–(2). Overall R^2.

Table A6: Earnings Returns: Robustness

	(1) Labor Income (IHS)	(2) Log Labor Income (Workers)	(3) Any Labor Income	(4) Labor Income Men	(5) Labor Income Women
Tertiary	4963.70***	0.398***	0.080***	4943.30***	4894.30***
	(413.20)	(0.038)	(0.011)	(494.60)	(778.40)
Non-White	−2313.70***	−0.247***	−0.041**	−2880.70***	−1161.10*
	(353.70)	(0.037)	(0.013)	(463.30)	(559.00)
Tertiary × Non-White	560.80	0.048	0.041*	848.80	−76.75
	(666.40)	(0.069)	(0.020)	(979.70)	(1025.70)
Tertiary effect, Non-White	5524.50***	0.446***	0.122***	5792.10***	4817.60***
Observations	25147	20401	25147	17207	7940
Individuals	3660	3485	3660	2337	1391
Overall R^2	0.24	0.22	0.14	0.20	0.22

Note: Source: PSID, 1999–2023. Random-effects estimates pooling White and Non-White individuals; standard errors clustered by individual. $^{+}$ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Column (2): workers only. Column (3): linear probability model. Tertiary effect, Non-White is the sum of Tertiary and Tertiary × Non-White. Controls as in Table 3.

A4 Birth-Cohort Differences

Table A7 compares the college wealth premium for two birth cohorts: individuals born in 1939–58 and those born in 1959–88. To compare people at a similar point in the life cycle, the sample is restricted to ages 40 to 50. Each column is a single model that pools White and Non-White individuals and interacts tertiary education with Non-White status and with an indicator for the later cohort. The lower panels of the table report the implied college premium for each race and cohort and the change between cohorts. The triple interaction tests whether that change differs by race.

Because of the age restriction, the two cohorts are observed in different years: the earlier cohort mainly in 1999–2009 and the later cohort mainly in 2001–2023. Differences between cohorts may therefore also reflect conditions in particular years, such as the Great Recession. Columns (2) and (4) restrict the sample to 2001–2009, when both cohorts are observed at ages 40–50, so that the two cohorts are compared in the same years.

Table A7: Cohort Interaction Tests

Dependent Variable: Wealth				
	Random Effects		**Within-Sibling**	
	All Years	2001–2009	All Years	2001–2009
Tertiary	6461.56***	6769.91***	6887.97	8909.88
	(1489.70)	(1778.55)	(7391.53)	(7489.09)
Tertiary × Non-White	1301.02	4059.79	−3682.91	−6153.10
	(3003.36)	(3033.21)	(8694.34)	(9073.97)
Tertiary × 1959–88	−1398.83	−3296.99	−2770.63	1004.27
	(1889.41)	(2454.08)	(7891.46)	(8716.55)
Tertiary × Non-White × 1959–88	−10909.23**	−4391.63	−1783.53	−4327.50
	(3729.02)	(4198.89)	(9417.95)	(10778.72)
Implied college wealth premium				
White, 1939–58	6461.56***	6769.91***	6887.97	8909.88
	(1489.70)	(1778.55)	(7391.53)	(7489.09)
White, 1959–88	5062.73***	3472.92+	4117.34	9914.15*
	(1285.49)	(1783.34)	(2738.72)	(4526.37)
Non-White, 1939–58	7762.59**	10829.70***	3205.06	2756.78
	(2547.31)	(2562.28)	(4628.74)	(5218.59)
Non-White, 1959–88	−4545.47*	3141.08	−1349.10	−566.46
	(1880.00)	(2413.47)	(2329.80)	(3807.98)
Cohort change in the college wealth premium				
White	−1398.83	−3296.99	−2770.63	1004.27
	(1889.41)	(2454.08)	(7891.46)	(8716.55)
Non-White	−12308.06***	−7688.62*	−4554.16	−3323.24
	(3162.02)	(3467.36)	(5195.36)	(6494.05)
Observations	8293	3464	3740	1910

Note: Source: PSID, 1999–2023, ages 40–50. Standard errors clustered by individual (random effects) or sibling pair (within-sibling). + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Each column is one pooled model interacting tertiary education with Non-White status and the 1959–88 cohort. Implied premia and cohort changes are linear combinations of these coefficients. Columns (2) and (4) use 2001–2009, when both cohorts are observed at ages 40–50. Controls as in Table 2, with the cohort indicator in place of birth-cohort effects.

A5 Robustness to the Treatment of Wealth

Table A8: Robustness: Alternative Wealth Measures

	White			Non-White		
	Log	Winsorized	Trimmed	Log	Winsorized	Trimmed
Panel A: Baseline random effects						
Tertiary	0.31***	3471.49***	3732.56***	0.64***	−3449.80***	−2474.97*
	(0.04)	(778.24)	(745.22)	(0.09)	(1040.37)	(1010.11)
Observations	13322	16259	15752	5704	8888	8795
R^2	0.19	0.24	0.25	0.10	0.11	0.10
Panel B: Within-sibling						
Tertiary		5595.16*	6152.62**		−660.52	−959.81
		(2199.53)	(2042.41)		(1827.26)	(1738.08)
Observations		4177	4047		3693	3675
R^2		0.03	0.03		0.01	0.01

Note: Source: PSID, 1999–2023. $^{+}$ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Specifications as in Table 2. Log: log of transformed wealth, positive wealth only (log of raw wealth: 0.61 White, 1.00 Non-White). Winsorized and trimmed at the 1st and 99th percentiles. No log model within siblings, since wealth differences can be negative.

A6 Parental Job Loss Instrumental Variables

Table A9: IV Estimates: Parental Job Loss

	Baseline		**+ Labor Income**		**+ Student Loans**	
	White	Non-White	White	Non-White	White	Non-White
Tertiary	26062.22*	78553.86	24474.60^{+}	82062.48	35114.35*	36520.76
	(12666.54)	(428657.50)	(12676.18)	(412615.50)	(15418.48)	(50786.21)
Labor income			0.20***	0.00		
			(0.06)	(0.92)		
Student loans					−0.97***	−1.27***
					(0.06)	(0.35)
First stage						
PJL	−0.17***	−0.01	−0.16***	−0.01	−0.18***	0.04
	(0.03)	(0.04)	(0.03)	(0.03)	(0.03)	(0.03)
F-statistic	31.03	0.06	30.47	0.07	32.00	2.07
Observations	8587	4756	8587	4756	4153	2550

Note: Source: PSID, 1999–2023 (student-loan columns: 2011–2023). Random-effects IV estimates. $^{+}$ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. PJL (parental job loss) is the instrument: an indicator for a parent losing a job when the child was aged 15–18. Year and cohort effects only. The instrument is weak for Non-White individuals; supplementary evidence only.